

READMe - Football League Implementation

Project Overview

This project is the design and implementation of a football league relational database using PostgreSQL. It covers the full development process, from data modelling through to physical implementation, including schema creation, data population, and query-based testing. The database is designed to accurately represent entities such as teams, players, fixtures, and events, while enforcing relationships and maintaining data integrity.

Database Summary

Key Database Elements:

- 10 Tables
- 40 Players
- 4 Teams
- 14 Fixtures
- 45 Fixture Events

Technologies Used

- **PostgreSQL** (pgAdmin 4)
 - Database Implementation & Testing
 - ER Diagram
- **Lucid Charts**
 - ER Diagram
- **Excel**
 - Data Preparation
 - CSV file conversion

Database Structure

Tables included in the database:

- **Groups** → Stores information about the groups players belong to
- **Teams** → Stores information about the teams players are assigned to
- **Matches** → Defines and stores information about different match types (i.e. Competitive or Friendly)
- **Venues** → Stores information about fixture locations
- **Events** → Defines and stores information about different fixture event types (i.e. Goal, Yellow Card, Red Card)
- **Fixtures** → Stores information about each fixture
- **Fixture_Teams** → Stores information about the home or away status of a team in a given fixture
- **Schedule** → Stores fixture date and time information
- **Players** → Stores information about each player
- **Fixture_Events** → Stores information about the events occurred in a fixture

Entity relationships were enforced using primary and foreign keys.

Design Decisions

Prior to implementation, the following improvements were made:

- Renamed 'Results' entity → 'Fixture_Events'
- Altered naming conventions → snake_case
- Introduced new 'Schedule' entity
- Renamed 'Date', 'Time' and 'Week' attributes → 'scheduled_date', 'scheduled_time', 'scheduled_week'

Database Implementation

1. Create and Connect to Database

Process:

- a. Connect to the desired server
- b. Right-click databases → Create → Database
- c. Name database 'football_league' → Save
- d. Right-click 'football_league' → Query Tool

2. Create Tables

Tables stored in the public schema under 'football_league'

Process:

- a. Run table creation script using the following script syntax

```
CREATE TABLE table_name (  
    column_name data type CONSTRAINT,  
    column_name data type CONSTRAINT,  
);
```

3. Populate Tables

Using the provided CSV files, import data into the created tables

Process:

- a. Right-click on a table under the schema tab → Import/Export Data
- b. Import → locate CSV file
- c. Options → check headers → set delimiter as "," → OK
- d. Check if data population was completed successfully using the following script syntax

```
SELECT *  
FROM table_name;
```

Testing

Verifying the functionality of the defined schema, the following SQL queries were run:

- List all players who belong to Cohort 7 and their respective teams
- List all fixtures that take place on August 10th, 2022, including team name and venue
- For each team, present the distribution of goals scored in each half of the match

Each test includes in-SQL documentation, SQL queries, and output results.

Future Work

This database can be expanded by:

- Adding further player information, e.g. age or height
- Adding further fixture information, e.g. status (scheduled, postponed, cancelled, etc.)
- Adding further venue information, e.g. (capacity, surface type)
- Adding a new 'Manager' entity with relation to 'Teams'
- Adding a new 'Referee' entity with relation to 'Fixtures'
- Adding a new 'Seasons' entity with relation to 'Schedule'